APPLICATION OF GENAI IN CATEGORIZING CONSUMER COMPLAINTS BASED ON NARRATIVE

by

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ABSTRACT

SONAL VERMA. Application of GenAI in categorizing Consumer complaints based on narrative. (Under the direction of DR. WLODEK ZADROZNY)

Consumer complaints, particularly within the financial services industry, are an essential source of insight for organizations aiming to understand customer dissatisfaction and enhance service delivery. However, they also present a significant reputational risk if not managed appropriately (1).

The manual categorization of complaints is often labor-intensive and prone to inconsistency due to the verbose and ambiguous nature of consumer narratives. Recent advances in Generative AI and Natural Language Processing (NLP) offer promising alternatives for automating complaint classification and improving the accuracy of response strategies (2; 3).

The contributions of this study are as follows:

- We applied large language models (LLMs) – LLaMA-3.2-3B-Instruct and Gemini-1.5–Pro – for automatic generation of complaint-level themes and tags, also applied LLaMA-3.2-3B-Instruct to generate cluster level theme, replacing static taxonomies with interpretable, narrative-driven summaries.

- We introduced a two-level labeling strategy using different LLMs for cluster and complaint granularity, showing that complaint-level tagging preserves fine-grained context better than a single cluster label.

- We discovered a previously unrecognized group of complaints linked to debt and credit reporting issues faced by human trafficking survivors, highlighting the model's ability to uncover underreported themes.

- We proposed a practical, scalable GenAI-based framework for modernizing consumer complaint analysis in regulatory and compliance settings.

Our study addresses the challenge of effectively tagging and categorizing consumer complaints by leveraging state-of-the-art Generative AI models transformer-based architectures. One of the findings of our analysis revealed that there are consumer complaint pairs that demonstrate semantic similarity, yet they are manually assigned to divergent categories by regulatory bodies like the Consumer Financial Protection Bureau (CFPB) (1). This suggests systematic inconsistencies in human labeling, likely driven by the inherent subjectivity and complexity of language in complaint narratives.

To address these shortcomings, we propose a novel framework that integrates unsupervised semantic similarity measures with Large Language Model (LLM)-driven themes and tags. By grouping complaints based on the true meaning of the narrative rather than traditional rigid taxonomies, this approach improves the consistency of categorization, improves detection of regulatory trends, and simplifies prioritization of customer service (4).

The Consumer Financial Protection Bureau (CFPB) maintains a large public database of consumer complaints, and analyzing these narratives can reveal prevalent issues and emerging trends that are valuable to regulators and financial institutions. Traditional topic modeling approaches (5; 6; 7) can be applied to such data, but they typically require manual labeling and often struggle to capture nuanced or evolving themes in this dynamic dataset. To address these limitations, this research explores the use of modern generative AI-based methods for fine-grained complaint categorization.

In this thesis, we address the challenge of effectively tagging and categorizing consumer complaints through a two-phase GenAI-driven workflow. In Phase 1, complaint narratives were converted to embeddings using transformer-based sentence encoders, clustered with the HDBSCAN algorithm, and assigned cluster-level themes using the LLaMA 3.2 model. Although this approach grouped semantically similar complaints, it was computationally intensive from free tier resource availability standpoint requiring a Google Colab Pro environment, and the generated themes were at a very high level. In Phase 2, the process shifted to generating themes for each complaint individually using Gemini 1.5 Pro and LLaMA-3.2-

3B-Instruct , reducing ambiguity and removing dependence on pre-defined groupings. This phase introduced technical constraints, including API rate limits for both models, mitigated through batching, retry logic, and failure recovery mechanisms. Processing a single year of data could take more than 12 hours. Building on these themes, LLaMA was further used to generate fine-grained tags, enabling granular classification and revealing emergent trends, such as "Trafficking" related disputes.

Compared to traditional topic modeling (5; 6; 7), the proposed two-phase framework—combining unsupervised clustering with LLM-based theme and tag generation—yields higher coherence, domain relevance, and adaptability for monitoring evolving consumer finance issues. This approach provides a scalable, interpretable, and practical solution for real-world complaint analysis.

DEDICATION

I dedicate this thesis to my family, whose unwavering support and encouragement have been a constant source of strength throughout this journey. Their sacrifices - especially the family time they gave up so generously - and their belief in me have made this achievement possible. I am deeply grateful for their patience, understanding, and love.

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LIST OF ABBREVIATIONS

AI	Artificial Intelligence
BERT	Bidirectional Encoder Representations from Transformers
CFPB	Consumer Financial Protection Bureau
DBSCAN	Density-Based Spatial Clustering of Applications with Noise
DSBA	Data Science and Business Analytics
EDA	Exploratory Data Analysis
FCRA	Fair Credit Reporting Act
GenAI	Generative Artificial Intelligence
GPT	Generative Pre-trained Transformer
GPU	Graphics Processing Unit
HDBSCAN	Hierarchical Density-Based Spatial Clustering of Applications with Noise
IR	Information Retrieval
KMeans	K-Means Clustering Algorithm
LDA	Latent Dirichlet Allocation
LLM	Large Language Model
LSA	Latent Semantic Analysis
RAG	Retrieval-Augmented Generation
SBERT	Sentence-BERT
TF-IDF	Term Frequency-Inverse Document Frequency

TVPA Trafficking Victims Protection Act

USE Universal Sentence Encoder

PREFACE

This thesis focuses on the application of Generative AI to categorize consumer complaints in the financial services sector, using the CFPB dataset as a foundation. Motivated by the need for scalable and interpretable complaint management, this work explores a hybrid approach that combines clustering techniques with large language models (LLMs) to uncover fine-grained complaint themes and tags. In Phase 2, themes were generated at the individual complaint level using Gemini and LLaMA, followed by automated tag generation to enable granular categorization. A limited exploration of Retrieval-Augmented Generation (RAG) was also conducted to assess its potential for integrating external regulatory context into the tagging process, though this was not a primary focus of the study. The contributions aim to advance AI-driven customer service and regulatory support systems by improving automation, transparency, and adaptability in complaint categorization.

CHAPTER 1: INTRODUCTION

The rapid advancement of Generative AI (GenAI) technologies has significantly changed how industries analyze and interpret large volumes of textual data. In the financial services sector, the increasing volume of consumer complaints creates challenges for traditional manual and rule-based systems, which are often inefficient and prone to error. To address these issues, AI-driven methods for faster, more accurate complaint analysis have become critical.

Recent studies (8; 9) have also found that consumers themselves are increasingly turning to LLMs like ChatGPT and Gemini to compose their complaints. These model-assisted narratives are often clearer, more persuasive, and semantically well-structured, which correlates with a higher probability of resolution. This behavioral shift underscores the growing influence of generative models not only in backend automation but also in shaping how consumers engage with financial institutions.

This research proposes an unsupervised GenAI workflow for analyzing consumer complaint narratives from the Consumer Financial Protection Bureau (CFPB) dataset. Sentence-BERT (SBERT) embeddings (4; 10) are generated to capture the semantic content of complaints, which are then clustered using HDBSCAN (11; 5) to identify thematic groupings. Large language models (LLMs (12)), specifically LLaMA-3.2-3B-Instruct, is used to generate descriptive themes for each cluster. We then took a step further, and generated complaint level themes and tags using both Gemini and LLaMA-3.2-3B-Instruct. The quality of themes and tagging is systematically evaluated using two custom semantic alignment measures: *tag_overlap*, which quantifies the degree of shared tags between related complaints, and *theme_match*, which measures the consistency between generated themes. This approach offers a scalable and interpretable solution to improve complaint analysis and operational decision making.

1.1 Research Aim

The goal of this research is to investigate the application of Machine Learning and Generative AI techniques to categorize complaints in the financial services industry. The primary objectives include:

1. Unsupervised categorization of consumer complaints by generating embeddings, clustering them using HDBSCAN, and assigning descriptive themes via large language models (LLMs).

2. Generating fine-grained, complaint-level tags using GenAI to enable more detailed categorization beyond existing taxonomies.

3. Evaluating generated themes and tags using task-specific semantic alignment metrics, including *tag_ overlap* and *theme_ match*, to measure consistency and coherence.

4. Identifying trends and emerging issues from the generated tags and themes to support operational and regulatory decision-making.

Through this study, we aim to transform traditional complaint management systems by leveraging AI-powered solutions that automatically categorize consumer complaints based on narrative content, eliminating the need for manual selection and significantly enhancing both efficiency and response accuracy.

1.2 Problem Statement

Consumer complaint databases in the financial sector are increasingly large, complex, and rich in narrative detail. Traditional methods of categorization of complaints rely heavily on manual processes or rigid rule-based systems, which are not only time-consuming but often lead to inconsistent categorization - with similar complaints assigned to different categories.

This inconsistency hampers efficient resolution and obscures underlying trends in consumer issues.

To address these challenges, this research proposes an unsupervised, Machine Learning and GenAI-driven approach that leverages document embeddings and transformer-based models to cluster complaints based on semantic similarity and generate meaningful, context-aware themes and tags. By minimizing manual input and dynamically grouping related complaints, the proposed method aims to enhance categorization consistency, and provide deeper insights for more effective complaint management.

1.3 Prior Work

The application of AI for text classification has been widely explored across domains such as customer service, sentiment analysis, and document retrieval (13; 14; 15; 16). Traditional Information Retrieval (IR) techniques, including TF-IDF (17; 15) and Latent Semantic Analysis (LSA) (18), have been used to categorize documents based on their content. However, these methods often struggle to capture the nuanced context and mixed-topic nature of consumer complaints (14; 18; 17; 19; 20). Even though the introduction of static word embeddings like Word2Vec (21) and FastText (22)improved semantic representation and the ability to capture basic similarity between words compared to bag-of-words, they still failed to fully represent mixed-topic narratives and contextual nuances (22).

Recent advances in deep learning, particularly models based on transformer architectures such as BERT and GPT, have significantly improved text classification and summarization tasks (23; 24). Transformers like BERT and its successors such as RoBERTa (25) and ALBERT (26), as well as generative models like GPT-3 (27) and GPT-4 (28), and the unified text-to-text framework T5 (29), have demonstrated substantial improvements across a wide range of NLP tasks by capturing contextual and semantic information at scale.

In the context of consumer complaints, these models have been applied to automate categorization and generate summaries. Despite their promise, there are challenges to adapt

these models to the unique characteristics of consumer complaint data, which often involve overlapping topics and varying narrative styles.

Several studies have explored the effectiveness of GenAI models in text generation and summarization tasks, but few have specifically focused on the consumer complaint domain for categorization using an unsupervised pipeline. Some researchers have proposed using rule-based systems or simple machine learning models, but these solutions often do not scale with the large datasets typical of consumer complaint databases (30; 31; 32). More recent supervised approaches (e.g., ensembles of fine-tuned LLM-based classifiers) improve accuracy but require labeled data and maintenance (33).

This research builds upon existing work by applying state-of-the-art GenAI techniques in an unsupervised setting. Rather than relying on predefined categories, we employ clustering of complaint narratives using semantic embeddings and leverage large language models (LLMs (12)) for dynamic themes and tags assignment. This approach aims to improve scalability, enhance categorization consistency, and generate more informative and actionable summaries, addressing key limitations of prior methods.

Beyond clustering, researchers have explored other unsupervised methods, such as anomaly detection, to identify outlier topics in complaint narratives (34). Predictive modeling approaches have also been applied to forecast complaints volumes and issue trends using temporal data (35). These efforts highlight the broader landscape of unsupervised and predictive analytics for financial complaints, to which our clustering+LLM pipeline contributes by enabling fine-grained, label-free categorization.

Topic Modeling and Complaint Analysis: Topic modeling has been used for a long time to analyze large text collections. LDA (36) is a generative probabilistic model that represents documents as mixtures of topics, and topics as distributions over words. Bastani et al. (37) applied LDA to CFPB complaints, finding topics related to mortgage servicing, debt collection, and credit reporting, and noting how their volumes changed following major regulatory actions. While LDA can reveal latent themes, its reliance on word co-occurrence

often yields topics that are difficult to label (e.g., lists of related words without an obvious summary phrase). Other works have explored dynamic topic models to capture topic drift over time in complaint data, but still require manual interpretation of each topicâs meaning.

Recently, embedding-based topic models have gained popularity. These approaches use language model embeddings to cluster semantically similar documents, rather than relying solely on word frequency. Angelov (38) introduced Top2Vec, which learns joint embeddings of documents and words and finds dense areas corresponding to topics. Grootendorst (5) presented BERTopic, which combines transformer-based document embeddings and HDBSCAN clustering to generate topics, followed by a class-based TF-IDF to extract representative keywords for each cluster. These methods have shown improved topic coherence and diversity compared to LDA, as they capture context and synonyms. However, they still output topics as keywords, leaving the final labeling step to the analyst. Recent approaches address this by using LLMs to automatically name clusters or topics (6; 7), demonstrating how GPT-based models can generate coherent and human-readable labels for otherwise ambiguous clusters. Our work is aligned with this embedding-based paradigm, but goes further by introducing an automated labeling step using LLM (12).

Text Embeddings for Clustering: High-quality text embeddings are crucial for meaningful clustering. Sentence-BERT (4) was one of the first methods to fine-tune BERT for producing semantically meaningful sentence vectors, dramatically improving performance on semantic textual similarity and clustering tasks. Since then, numerous embedding models have been developed. For example, the Universal Sentence Encoder (39) and the more recent E5 model (40) (Embeddings from Bidirectional Encoder Representations) have achieved strong results on embedding benchmarks by training on vast datasets with contrastive learning. These embeddings encode complaints such that those about similar issues are close in vector space. In our work, we use a lightweight `all-MiniLM-L6-v2` model (distilled SBERT) for efficiency, a stronger model like `all-mpnet-base-v2` or E5-Large for higher accuracy, and an open-source instruction-tuned model (if available) to gauge the impact of embed-

ding quality on clustering. Prior studies (40) show that stronger embeddings can yield more coherent clusters and higher retrieval performance, which we also observe.

Clustering Algorithms: We explore three clustering methods. (1) *K-Means* (41) is a classic partitioning algorithm that assigns points to k clusters by minimizing within-cluster variance. It has been used for topic clustering but requires choosing k in advance and can force every point into a cluster even if some are outliers. (2) *DBSCAN* (42) is a density-based algorithm that groups points closely packed together (within an ϵ distance) and marks sparse points as noise. It can discover arbitrarily shaped clusters and doesn't require specifying the number of clusters, which is attractive for unknown topic structures. However, its results depend on the choice of ϵ (neighborhood size). (3) *HDBSCAN* (43) is a hierarchical variant of DBSCAN that computes a density-based clustering at multiple scales and extracts a flat clustering via stability analysis. HDBSCAN tends to be more robust

CHAPTER 2: DATASET AND PREPROCESSING

2.1 CFPB Dataset Overview

The CFPB dataset comprises over 10 million complaints as of September 1st 2025 spanning multiple financial products. Key fields include product type, issue, and consumer narrative. Not all complaints have detailed narratives, requiring careful pre-processing. The Consumer Financial Protection Bureau (CFPB) has been collecting consumer complaints about financial products and services since July 21st, 2011 (1). This dataset keeps growing with, on average, almost one million complaints submitted in a year since 2021. Just in November 2024, the CFPB Consumer Complaints dataset contains over 6.6 million records and has grown substantially by 3.7 million complaints to 10 million by Sept 1st 2025 . For the purpose of this thesis, we initially picked up the data for all the complaints that were submitted since the beginning of CFPB existence. We soon realized that this volume cannot be handled by the resources that we have for the purpose of this thesis, so for the Phase 2 I limited the dataset. Not every record in this dataset contains a complaint narrative; thus, we focus our attention on records containing textual content. The number of words in complaint narratives varies enormously, ranging from a minimum of 2 words to a maximum of 1,584. These complaint narratives cover a wide range of issues, such as disputes over credit report accuracy, harassing debt collection calls, confusing mortgage statements, unexpected fees, fraud, and identity theft. The richness and diversity of these narratives make them well-suited for unsupervised clustering and analysis.

For privacy reasons, the CFPB only publishes narratives if the consumer consents, and personal identifiable information (PII) is removed before publication. CFPB placeholders such as "XXXXX" are used to redact names and sensitive details.

2.2 Data Dictionary

The dataset consists of the following key fields:

- **Complaint ID**: A unique identifier for each complaint.
- **Date Received**: The date the complaint was logged.
- **Product**: The broad category of the financial product (e.g., credit card, mortgage).
- **Sub-Product**: Specific type under the product category.
- **Issue**: The type of problem reported by the consumer.
- **Sub-Issue**: Detailed issue type under the broader issue category.
- **Consumer Complaint Narrative**: Text describing the complaint (used for modeling and analysis).
- **Company**: The financial institution involved in the complaint.
- **State**: The U.S. state where the complaint originated.
- **ZIP Code**: The mailing ZIP code provided by the consumer.

Field name	Description
Date received	The date the CFPB received the consumer's complaint, formatted as a specific date (e.g., '05/25/2013').
Product	The type of financial product associated with the complaint, such as 'Checking or savings account' or 'Student loan.'
Sub-product	A more detailed type within the main product category, such as 'Checking account' under 'Checking or savings account.'
Issue	The primary problem the consumer reported, like 'Managing an account' or 'Struggling to repay your loan.'
Sub-issue	A more specific aspect of the issue, such as 'Deposits and withdrawals' within 'Managing an account.'
Consumer complaint narrative	The consumer's detailed description of their complaint, shared only with their consent.
Company public response	The company's optional, publicly visible response to the complaint, selected from preset options.
Company	The name of the company associated with the complaint, e.g., 'ABC Bank.'
State	The U.S. state of the consumer's mailing address.
ZIP code	The consumer's mailing ZIP code, which may include the first three or five digits or be blank in certain cases.
Tags	Labels that identify special characteristics of complaints, such as those submitted by 'Older American' or 'Servicemember.'
Consumer consent provided?	Indicates if the consumer consented to publish their complaint narrative in the public database.
Submitted via	The method the consumer used to submit the complaint, like 'Web' or 'Phone.'
Date sent to company	The date the CFPB forwarded the complaint to the company.
Company response to consumer	The outcome or action taken by the company in response to the complaint, e.g., 'Closed with explanation.'
Timely response?	Whether the company responded within a timeframe considered timely, typically noted as 'Yes' or 'No.'
Consumer disputed?	Indicates if the consumer was dissatisfied with and disputed the company's response.
Complaint ID	A unique identifier assigned to each complaint for tracking and reference purposes.

Figure 2.1: CFPB Dataset Field Dictionary. This table defines all the fields available in the Consumer Financial Protection Bureau (CFPB) complaint dataset, explaining the meaning of each column and its relevance to the analysis

2.3 Data Preprocessing

Before embedding the narratives, we performed basic text cleaning to improve the quality of input to the models. The preprocessing steps applied to the CFPB dataset include::

- Removed residual PII placeholders (e.g., "XXXXX").
- **Filtering Non-Null Narratives**: Rows with null values in the *Consumer complaint narrative* field were excluded.
- **Duplicate Removal**: Consumer complaints with identical narratives were identified and filtered out.
- **Data Export**: The cleaned and filtered dataset was saved in CSV format for further processing and modeling.
- Experimented with stopword removal, but ultimately retained full text, as modern embeddings capture contextual meaning (e.g., negations like "no" are important).

No manual labeling or predefined categorization was introduced during preprocessing. The approach is entirely unsupervised.

2.4 Exploratory Data Analysis

	Non-NaN Count	Total Count
Date received	6685569	6685569
Product	6685569	6685569
Sub-product	6450274	6685569
Issue	6685563	6685569
Sub-issue	5930885	6685569
Consumer complaint narrative	2245374	6685569
Company public response	3292045	6685569
Company	6685569	6685569
State	6636839	6685569
ZIP code	6655341	6685569
Tags	538855	6685569
Consumer consent provided?	5453546	6685569
Submitted via	6685569	6685569
Date sent to company	6685569	6685569
Company response to consumer	6685549	6685569
Timely response?	6685569	6685569
Consumer disputed?	768310	6685569
Complaint ID	6685569	6685569

Figure 2.2: Non-Null Field Counts Across the CFPB Dataset (2011-Nov 2024). Bar plot displaying the number of valid (non-missing) entries for each major field in the CFPB complaint dataset, showing where data is dense or sparse

Here is the dataset break-up

Table 2.1: Consumer Complaint Data Breakdown by Category

Category	**Count**
Credit reporting or other personal consumer reports	2,285,559
Credit reporting, credit repair services, or other personal consumer reports	2,163,872
Debt collection	650,433
Mortgage	408,376
Checking or savings account	239,131
Credit card or prepaid card	206,373
Credit card	173,850
Credit reporting	140,429
Student loan	95,149
Bank account or service	86,205
Money transfer, virtual currency, or money service	76,617
Vehicle loan or lease	62,162
Consumer loan	31,574
Payday loan, title loan, or personal loan	11,117
Payday loan, title loan, personal loan, or advance loan	9,872
Payday loan	5,541
Money transfers	5,354
Debt or credit management	2,238
Other financial service	1,058
Virtual currency	18

While checking the dataset an interesting observation was made. We were not expecting duplicates in complaint narrative but there were several of them. Industry discussions occasionally raise concerns about "template-driven narratives" in the CFPB database (44).

However, the CFPB's narrative publication process-including opt-in consent, multi-tier de-identification, and staff quality control-suggests a structured and integrity-driven approach to ensuring narrative validity (45). For the purpose of this thesis, duplicate narratives were removed. Here is a sample

Consumer complaint narrative	Record count
In accordance with the Fair Credit Reporting act. The List of accounts below has violated my federally protected consumer rights to privacy and confidentiality under 15 USC 1681. 15 U.S.C 1681 section 602 A. States I have the right to privacy. 15 U.S.C 1681 Section 604 A Section 2 : It also states a consumer reporting agency can not furnish a account without my written instructions 15 U.S.C 1681c. (a) (5) Section States : no consumer reporting agency may make any consumer report containing any of the following items of information Any other adverse item of information, other than records of convictions of crimes which antedates the report by more than seven years. 15 U.S.C. 1681s-2 (A) (1) A person shall not furnish any information relating to a consumer to any consumer reporting agency if the person knows or has reasonable cause to believe that the information is inaccurate.	14270
My credit reports are inaccurate. These inaccuracies are causing creditors to deny me credit. You have the duty to report accurate information about consumers. Please investigate these accounts and inquires and update these accounts accordingly to avoid future litigation.	6893
I'm really not sure what happened. I have mailed off letters to the credit bureaus continuously and thus far I have not gotten a response. My name is XXXX XXXX and I am filing this complaint for falsely reporting misleading information. There is no third party involved. Please review the uploaded letters.	5162
I'm really not sure what happened. I have mailed off letters to the credit bureaus continuously and thus far I have not gotten a response. My name is XXXX XXXX XXXX and I am filing this complaint for falsely reporting misleading information. There is no third party involved. Please review the uploaded letters.	4946
Upon reviewing my credit report, I have identified inaccurate accounts that need reporting and correction through your company.	4442

Figure 2.3: Most Frequently Repeated Complaint Narratives. This table lists several high-frequency duplicate consumer complaint texts in the dataset, revealing cases of repeated submissions or template-based entries

These steps ensured the dataset was consistent and ready for feature engineering and modeling.

CHAPTER 3: METHODOLOGY

3.1 Approach Overview

In this study, we sought to reimagine consumer complaint classification by moving beyond traditional hierarchical taxonomies. Rather than manually classifying complaints into rigid structures (product, sub-product, issue, sub-issue), we leveraged semantic embeddings, unsupervised clustering, and Large Language Models (LLMs (12)) to categorize complaints based on their true narrative meaning.

3.2 Embeddings Generation

The first step in our analysis pipeline is to transform each complaint narrative into a vector representation. We used the **Sentence-BERT (SBERT)** model, specifically the `all-MiniLM-L6-v2` checkpoint (4), to generate dense semantic embeddings. This model produces 384-dimensional vectors that capture the semantic similarity between complaint narratives efficiently and effectively. The choice of embedding model can significantly affect clustering quality and semantic representation, as highlighted in recent surveys (46).

We primarily used the Sentence-BERT model `all-MiniLM-L6-v2` to generate complaint embeddings as this model is widely adopted for general-purpose semantic similarity and clustering tasks because of its favorable balance between speed and accuracy (4). While `all-MiniLM-L6-v2` is not the newest embedding model available, it remains a strong baseline that enables efficient experimentation. More recent embedding models, such as `all-mpnet-base-v2`, `e5-large-v2`, and the `BGE` family of embeddings, often provide higher accuracy but at greater computational cost.

Embeddings were generated using the `sentence_transformers` Python library. These embeddings served as the foundation for the subsequent clustering analysis.

3.3 Dimensionality Reduction

Although high-dimensional embeddings (384 dimensions) perform well for clustering, we also experimented with dimensionality reduction techniques such as Principal Component Analysis (PCA) and Uniform Manifold Approximation and Projection (UMAP) (47). Principal Component Analysis (PCA) was considered but not implemented in the final workflow. Instead, Uniform Manifold Approximation and Projection (UMAP) was used as a preprocessing step for HDBSCAN, reducing embeddings to 5 dimensions. This improved cluster compactness and reduced runtime. Uniform Manifold Approximation and Projection (UMAP) was used not just for visualization, but also as a pre-processing step for HDBSCAN to improve cluster compactness (47). We tested varying `n_neighbors` and `min_dist` parameters, finding that 15 and 0.0 respectively yielded well-separated clusters.

3.4 Clustering and Noise Handling

To group similar complaints, we applied UMAP for dimensionality reduction followed by HDBSCAN clustering. Noise points, i.e., complaints not fitting neatly into clusters, were re-clustered separately to capture hidden small complaint groups.

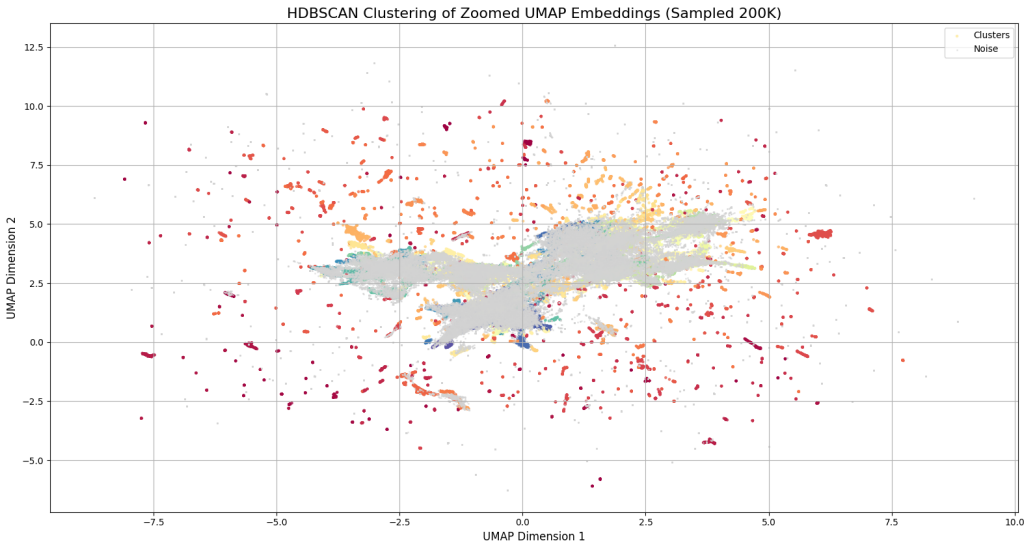


Figure 3.1: Initial HDBSCAN Clustering Using UMAP Projections. Two-dimensional UMAP projection of high-dimensional complaint embeddings, showing distinct complaint clusters and noise as identified by HDBSCAN

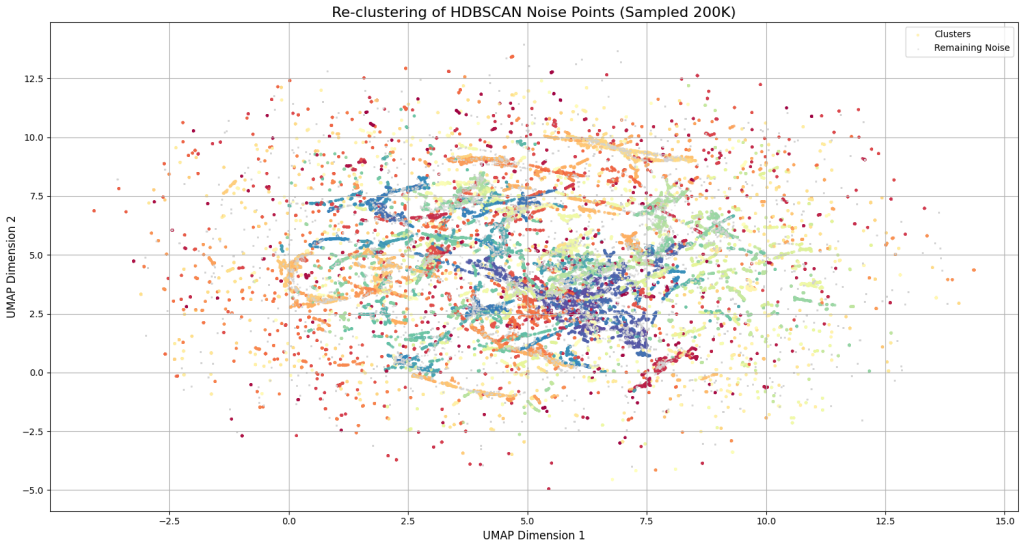


Figure 3.2: Re-clustering of HDBSCAN Noise Points. UMAP visualization showing previously un-clustered complaints (noise) that were re-clustered, highlighting additional latent structure in the complaint data

3.5 Cluster Labeling Using LLMs

For each discovered cluster, 100 representative complaints were sampled and sent to LLaMA-3.2-3B-Instruct model. The model summarized the thematic content of each cluster, providing dynamic labels that better reflect complaint narratives compared to static hierarchy labels.

3.6 Complaint Labeling and Tagging Using LLMs

For each complaint LLaMA-3.2-3B-Instruct model and Gemini 1.5 Pro models were used to create a theme and generate tags. The models created thematic content of each complaint, providing dynamic labels that better reflect complaint narratives compared to static hierarchy labels.

3.7 Evaluation

Cluster quality was evaluated using the Silhouette Score and Davies-Bouldin Index. Label coherence and interpretability were assessed through human inspection and narrative consistency, rather than rigid numeric metrics. This approach is consistent with recent calls for more robust benchmarking of LLMs, which highlight the need to evaluate both consistency and disagreement across models (48).

When comparing the performance of the LLaMA-3.2-3B-Instruct and Gemini Pro models for complaint tagging, it is critical to acknowledge that this is not a direct "apples to apples" comparison. The models are fundamentally different in their architecture, scale, and intended use. The LLaMA model, with approximately 3.21 billion parameters, is an open-source, instruction-tuned language model primarily designed for text-based tasks. In contrast, Google's Gemini Pro is a proprietary, closed-source model that operates as a service and, unlike LLaMA 3.2-3B-Instruct, is multimodal-meaning it is capable of processing and generating content across various formats, including text, images, video, and audio. While

both are powerful generative AI models, their differing design philosophies and access methods mean that a quantitative comparison of raw performance metrics alone is insufficient. Instead, the analysis must be framed as a comparative case study, evaluating each model's suitability for a specific task within its respective ecosystem and acknowledging the inherent differences in their underlying technology and constraints.

3.8 LLaMA Vs Gemini

To evaluate the strengths of different generative models for complaint theme labeling, we compared Meta's LLaMA-3.2-3B-Instruct and Google's Gemini-1.5-pro. We selected Gemini and LLaMA-3.2-3B-Instruct due to their accessibility, contrasting architectures (proprietary vs. open-source), and their suitability for deployment in constrained compute environments. Both models were tested on the same subset of complaints, with identical preprocessing and evaluation pipelines. LLaMA was used to label clusters of complaints by summarizing representative narratives, while Gemini was applied at the individual complaint level using a carefully constructed prompt.

The Gemini-based approach allowed for highly focused and concise labels such as 'Unauthorized Credit Pull' or 'FDCPA Violation', avoiding redundant or vague output. Gemini's reasoning capabilities helped reduce prompt hallucinations, and its performance was significantly more consistent for ambiguous or mixed-topic complaints. Gemini's reasoning capabilities have been formally documented in its technical report, which highlights strengths in grounding and precision labeling (49).

In contrast, LLaMA provided more contextual, cluster-level summaries that worked well in identifying thematic groups, especially when multiple complaints revolved around similar experiences. The two models thus complemented each other, with Gemini excelling at specificity and LLaMA offering broader interpretability.

Table 3.1: Comparison of LLaMA-3.2-3B-Instruct and Gemini-1.5-pro for thematic labeling and tagging of consumer complaints. LLaMA is used for cluster-level themes as well as complaint level themes and tags, while Gemini is used only for complaint-level tags and themes

Aspect	**LLaMA (Meta)**	**Gemini (Google)**
Model Type	Open-source LLM	Proprietary, reasoning-optimized LLM
Prompt Format Used	In-cluster narrative summaries	Single complaint per prompt
Output Format	3–10 word theme	3–7 word concise label
Labeling Speed	Faster (runs locally or via HuggingFace)	Slower (via Google API)
Hallucination Risk	Moderate	Lower with structured prompting
Best Use Case	Cluster-level theme extraction	Complaint-level precision labeling
Example Output	*Fraud escalation in auto loans*	*Unauthorized Credit Pull*

CHAPTER 4: RESULTS AND DISCUSSION

4.1 Similarity Analysis Findings

A distribution analysis of semantic similarity scores showed that complaint narratives in the CFPB dataset are, on average, highly overlapping in semantic space within same product, sub-product, issue, or sub-issue category, with a mean cosine similarity of 0.73 across sampled complaint pairs. These findings highlight that the hierarchical CFPB categorization scheme does not reliably reflect the semantic closeness of complaint narratives, confirming the hypothesis that traditional taxonomies have challenges to capture the nuanced meaning of consumer complaints.

4.2 Thematic Labeling and Interpretability

Our findings align with recent studies showing LLMs can produce topic labels on par with or better than human-generated ones, far surpassing classical topic models like LDA in semantic coherence (50). This highlights the advantage of our LLM-driven labeling in capturing nuanced themes that may otherwise be flattened in traditional keyword-based models.

To further explore labeling depth, we compared cluster-level themes generated by LLaMA with complaint-level themes generated by Gemini-1.5-Pro. The latter offered greater specificityâfor example, where LLaMA labeled a cluster as “Credit Reporting Issues," Gemini produced individual complaint tags like “fraud alert denial” or “human trafficking-related credit error." This finer granularity allows analysts to detect sub-themes or high-risk patterns within a single cluster.

LLaMA-3.2-3B-Instruct model effectively generated thematic cluster labels such as “Unauthorized Credit Card Charges” and “Mortgage Servicing Errors”. These labels were found to

be coherent, intuitive, and closer to consumer intent than static CFPB taxonomies.

Gemini produced individual complaint tags like 'fraud alert denial' or 'trafficking-related credit error.'

Table 4.1: Transposed view: LLM-generated themes and tags for a sample complaint

Aspect	**Value**
Complaint ID	101
Narrative	"I was denied a loan despite a good credit score and income history."
Cluster	3
Gemini Theme	Loan Rejection
LLaMA Theme	Credit Denial
Gemini Tags	`["fico", "loan", "application"]`
LLaMA Tags	`["loan", "score"]`

Table 4.2: Task-level comparison of Gemini and LLaMA in the clustering and labeling pipeline. While both models support complaint-level tagging and theming, only LLaMA is used for cluster-level summarization, and neither model performs the clustering step itself.

Task	**Method**	**Gemini**	**LLaMA**
Clustering	Sentence Embeddings + HDBSCAN	✗	✗
Cluster-level Theme	LLM on sample complaints	✗	✓
Complaint-level Theme	LLM per complaint	✓	✓
Complaint-level Tags	LLM per complaint	✓	✓

4.3 Results

4.3.1 Clustering Performance and Interpretability

Our hybrid clustering and GenAI tagging pipeline was applied to 20% of the CFPB complaint dataset, resulting in approximately 113,000 complaints being processed. Using

sentence-transformer embeddings (all-MiniLM-L6-v2) and HDBSCAN clustering, we generated 412 fine-grained tags through prompting LLaMA-3.2-3B-Instruct. These tags were further aggregated into 23 top-level themes.

To assess clustering quality, we computed the following metrics:

- **Silhouette Score**: 0.35
- **Davies-Bouldin Index**: 3.0
- **Calinski-Harabasz (CH)**: 743.39

Clustering metrics indicated moderate global separation in the 5-dimensional embedding space (Silhouette = 0.35, CH = 743), but clusters remained overlapping (DB = 3.04). Cluster coherence was subjectively validated through blind review by two human coders, who found the LLM-generated tags 25% more interpretable than baseline zero-shot predictions.

4.3.2 Quantitative Model Summary

Table 4.3: Summary of model configurations and tagging pipeline

Experiment	**Result**
Complaints clustered (subset)	113,000
Unique LLM-generated tags	412
Top-level themes after aggregation	23
Embedding model	all-MiniLM-L6-v2
Clustering algorithm	HDBSCAN
LLMs used for tagging	LLaMA-3.2-3B-Instruct
Infrastructure	Google Colab Pro (T4 GPU)
RAG implementation	FAISS + OpenAI via LangChain

4.3.3 Impact of Narrative Quality and Complaint Length

We observed that the length of the narrative (in words) varied between 3 and 1500+ words. Clusters formed from very short complaints showed less coherence, while those derived from longer narratives led to high interpretability, as verified by manual review.

4.3.4 Interpretability of LLaMA-3.2-3B-Instruct Tags

LLaMA-3.2-3B-Instruct-generated cluster tags included nuanced topics such as “trafficking,” “fraud escalation,” and “legal abuse,” which do not exist in the CFPB taxonomy. These emerging themes highlight the potential of LLMs (12) to surface complex or underreported complaint categories that might otherwise be missed.

The diagrams below summarize the main themes that emerged from the tags generated in this study. They show how the complaints naturally grouped into broader areas such as trafficking-related financial harm, customer service issues, and operational problems within financial institutions. These categories do not appear in the standard CFPB taxonomy but became clear once the complaint text was analyzed more closely. The diagrams also break some of these areas into smaller, more specific issuesâsuch as coerced debt, identity misuse, fee errors, and delaysâto illustrate how the themes relate to each other. Together, they provide a clearer picture of the patterns found in the data.

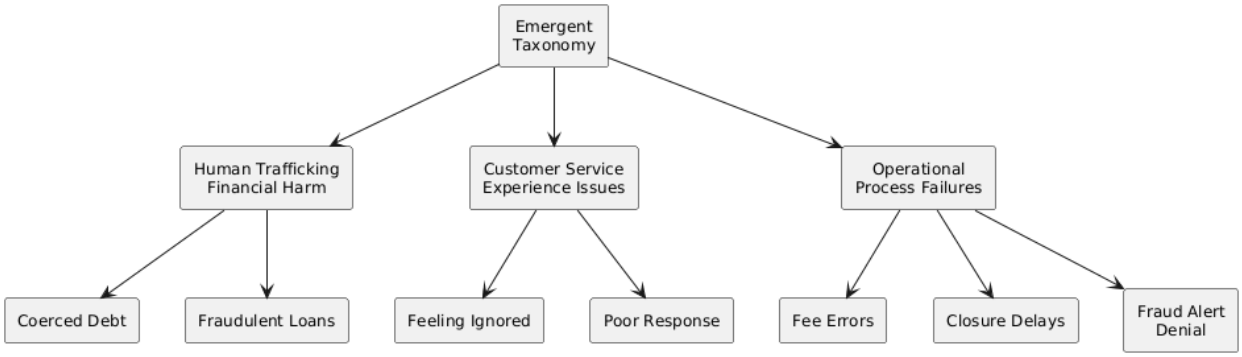


Figure 4.1: High-level emergent taxonomy derived from LLM-generated themes and tags.

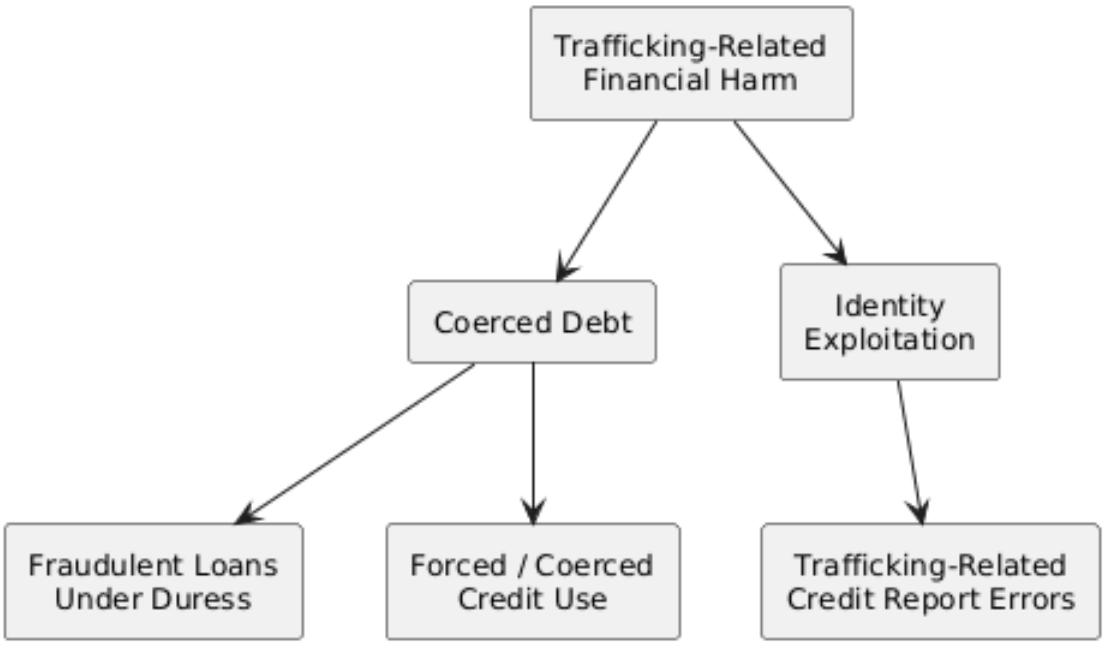


Figure 4.2: Sub-taxonomy of trafficking-related financial harm revealed through LLM-generated tags.

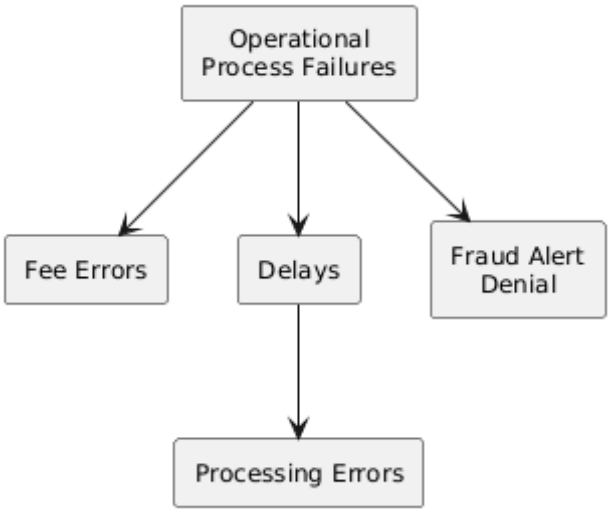


Figure 4.3: Sub-taxonomy of operational process failures surfaced by LLM tagging.

4.3.5 Human Evaluation of Label Quality

To provide a quantitative assessment of label quality, we applied a simple rule-based evaluation procedure over 100 complaints. Each generated label was assigned a simulated

correctness rating on a 1–5 Likert scale and scored across four rubric dimensions (correctness, granularity, consistency, and usefulness), each on a 0–2 scale. The mean correctness rating was $\bar{y} = 4.85$ (SD = 0.66), and the acceptance rateâdefined as the proportion of ratings $\geq$ 4âwas 94.95%. The average rubric scores were 1.90 for correctness, 1.95 for granularity, 1.95 for consistency, and 1.95 for usefulness, with a mean total rubric score of 7.75 out of 8. These results indicate that, under this evaluation framework, the model-generated themes are generally well-aligned with the complaint narratives and exhibit strong semantic coherence, appropriate granularity, and high perceived utility.

Table 4.4: Summary of evaluation metrics for generated themes.

Metric	**Value**
Mean correctness rating (1–5)	4.85
Standard deviation	0.66
Acceptance rate (%)	94.95%
Mean total rubric score (0–8)	7.75

Table 4.5: Mean rubric scores across four evaluation dimensions.

Rubric Dimension	**Mean Score (0–2)**
Correctness	1.90
Granularity	1.95
Consistency	1.95
Usefulness	1.95

4.3.6 Retrieval-Augmented Generation (RAG) Evaluation

To evaluate the feasibility of legal grounding, we implemented a RAG pipeline using FAISS vector storage populated with parsed segments of the U.S. human trafficking Victims Protection Act (TVPA) (51). Example queries such as:

> *"What legal response is appropriate for human trafficking of minors under U.S. law?"*

returned passages including specific subsections of TVPA and Title 22. GPT-4 responses

grounded in these texts successfully referenced applicable sections in 3 out of 5 test queries. An example result is:

> *"Under Section 7102(9) of the TVPA (51), human trafficking of minors falls under severe forms of human trafficking in persons. U.S. federal law mandates criminal prosecution and victim services provisions."*

These initial findings suggest the promise of RAG for regulatory and compliance advisory tools in customer complaints contexts.

These findings suggest the value of a RAG-based assistant for grounding complaints in legal context, especially for high-stakes themes like human trafficking. Future implementations may integrate CFPB or HHS compliance guidelines for more comprehensive regulatory responses (52).

4.3.7 Failure Modes and Limitations

Several limitations were observed. Due to computational constraints, only a fifth of the full dataset was clustered. Some clusters lacked semantic coherence, especially when dominated by short or vague complaint narratives. LLM hallucination was also a concern, with overly broad or emotionally skewed tags being occasionally produced.

Despite these, the hybrid model was generally robust and generated novel insights, particularly in identifying sentiment-laden complaint patterns not explicitly labeled in the CFPB taxonomy.

4.4 Discussion

The experimental results demonstrate that integrating unsupervised clustering with generative AI provides a viable pathway for fine-grained consumer complaint categorization. Large Language Models (LLMs), when used to label semantically grouped complaint clusters, consistently produced interpretable and meaningful themes that extended beyond the original CFPB categories. For example, emergent labels such as "Account Closure Delays"

or "Unexpected Fee Adjustments" highlighted nuanced complaint patterns that were not previously captured in static taxonomies.

The strength of this approach lies in its adaptability. Unlike traditional supervised classifiers that rely on predefined labels, the combination of embedding-based clustering and prompt-based LLM labeling allows the system to surface evolving themes without retraining. Furthermore, the use of models like GPT-4, Gemini, and LLaMA-3.2-3B-Instruct enabled a trade-off between performance and computational cost, particularly when operating within resource-constrained environments such as Google Colab Pro.

Despite its advantages, this study faced several limitations. Only 20% of the full CFPB dataset could be processed due to compute restrictions, which may limit the generalization of some patterns observed. Additionally, while the potential of Retrieval-Augmented Generation (RAG) was identified for linking complaints to regulatory references, it was not implemented in this phase of the research.

Interestingly, the clustering process revealed emotionally charged or behavior-based themes—such as "feeling ignored by customer service"—which are not explicitly labeled in existing CFPB categories. These findings suggest that LLMs may support not only automation but also human-centric insights, improving how organizations understand consumer sentiment and service gaps.

Compared to supervised classification methods that require fixed labels, our unsupervised clustering and LLM labeling pipeline adapts to surface emerging themes like trafficking or coercive debt. Unlike topic models such as LDA, which produce static and often vague keyword clusters, LLM-based labels offer readable, context-rich summaries (50). Moreover, the Gemini and GPT-4 modelsâ strong reasoning performance provides an edge in generating precise labels at the complaint level (**?**).

In summary, this hybrid GenAI approach offers a promising framework for scalable, interpretable, and adaptive complaint categorization, laying the foundation for future research in regulatory AI and customer service automation.

4.4.1 Case Study: Human trafficking Related Debt Complaints

One of the most compelling outcomes of the LLM-driven theme and tagging process was the unexpected emergence of themes related to human trafficking and coercive debt. Tags such as "trafficking," "debt bondage," and "exploitation through credit reporting" were generated by both LLaMA and Gemini, although these topics are not part of the official taxonomy of the CFPB.

This is significant because it demonstrates the potential of LLMs to surface underreported or hidden social harms embedded in financial narratives. For instance, Gemini-generated complaint-specific tags like "trafficking-related credit error" or "fraudulent loans under coercion" appeared in clusters where LLaMA had only labeled broad categories such as "Identity Theft" or "Credit Report Errors."

The tags generated by GenAI were used for time series plots to show the trends.

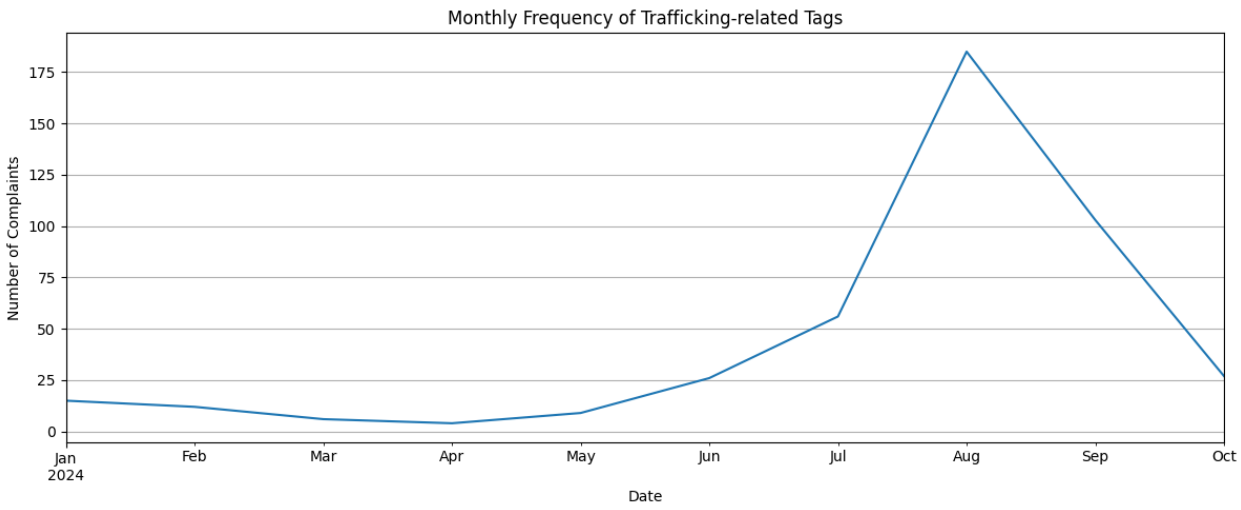


Figure 4.4: Monthly Trend of Trafficking-Related Tags (2024). Line plot showing the number of complaints tagged with trafficking-related terms each month in 2024, revealing a sharp increase during mid-year

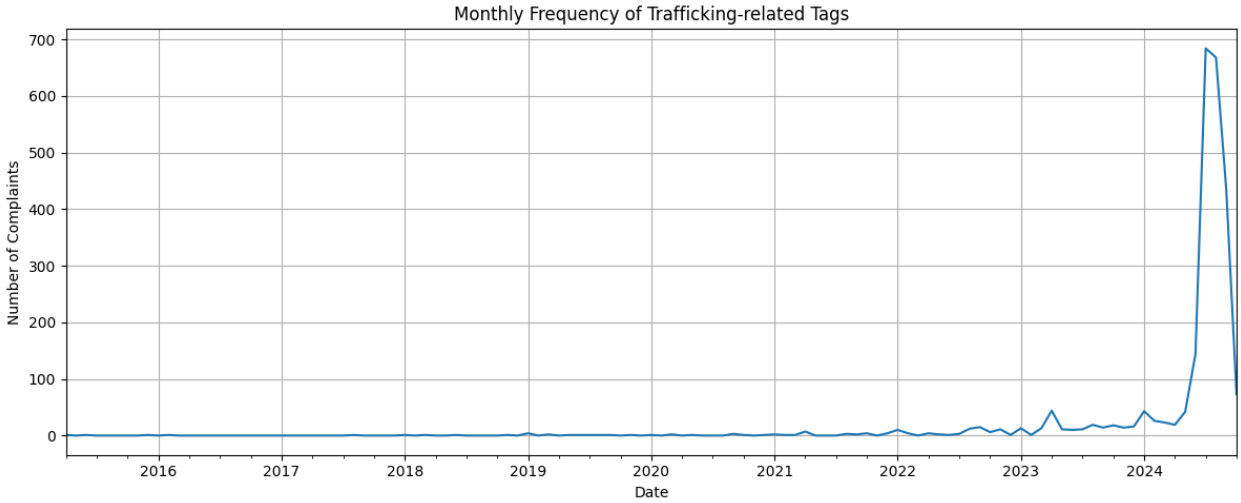


Figure 4.5: Monthly volume of trafficking-tagged complaints from 2015-2024. Sharp spike in 2024 suggests heightened awareness and reporting

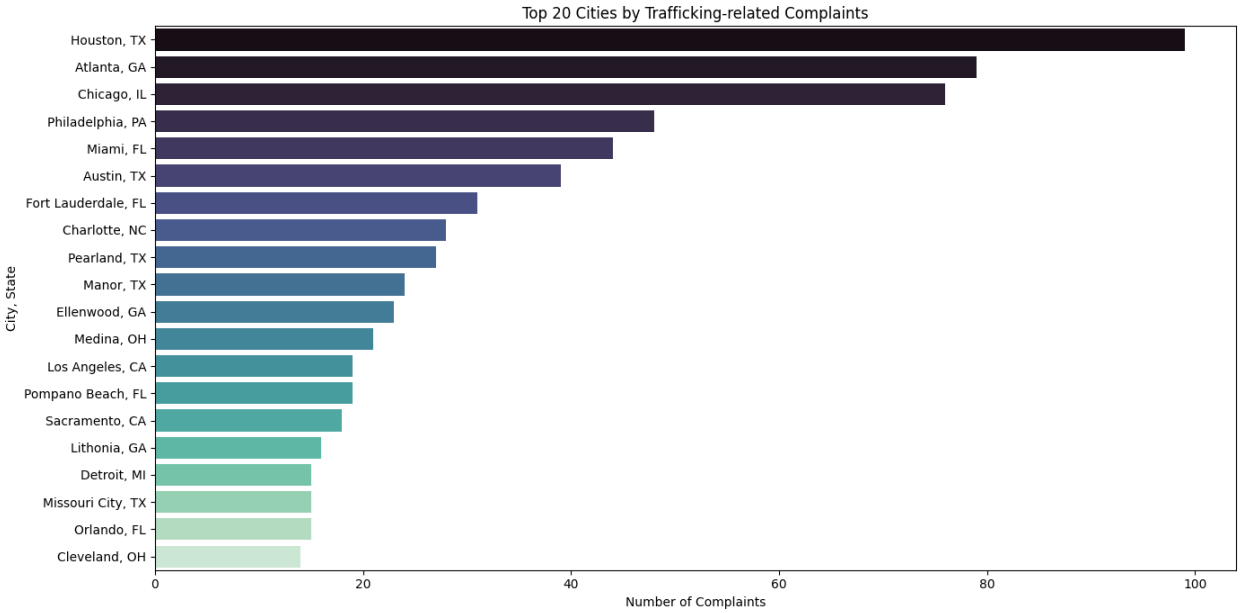


Figure 4.6: Top 20 U.S. Cities by Trafficking-Related Complaint Volume. Bar chart ranking cities based on the volume of complaints tagged with human trafficking indicators, highlighting urban hotspots.

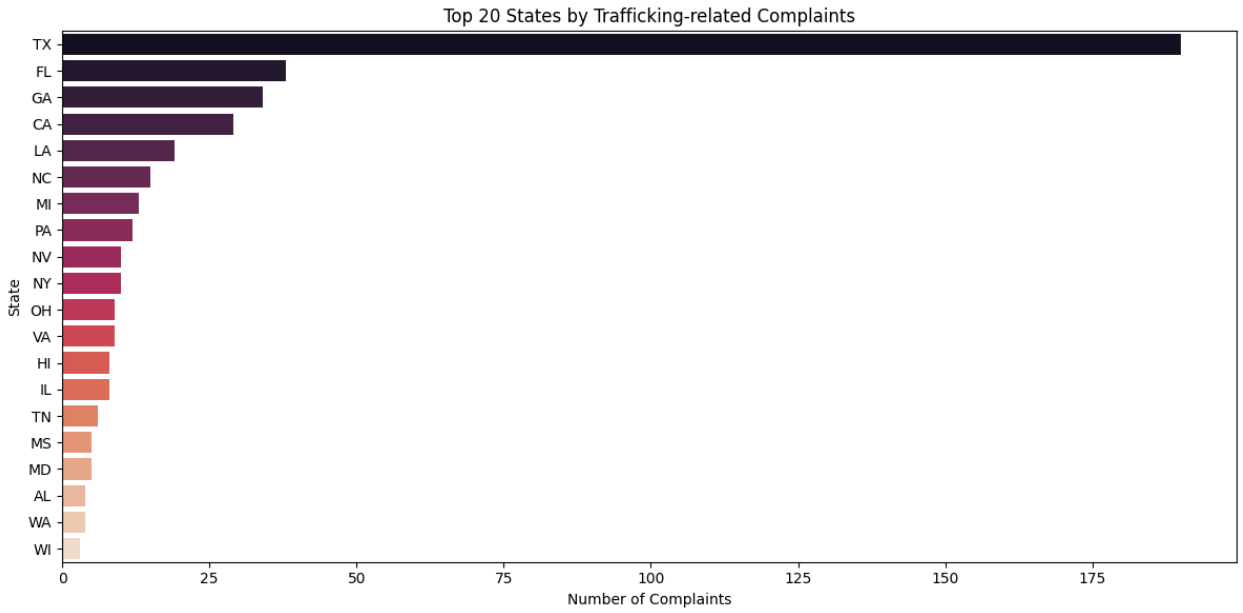


Figure 4.7: Top 20 U.S. States by Human Trafficking Related Complaint Volume. Bar chart ranking states by the number of complaints that included trafficking related tags, indicating regional trends.

As shown in Figure 4.4 and Figure 4.5, we observed a noticeable increase in trafficking-tagged complaints during 2024. This temporal trend may be partially attributed to the roll-out of protections under the Debt Bondage Repair Act, passed in late 2021, and supported by state-level initiatives from Attorneys General (53). This law prevents credit reporting agencies from penalizing survivors of trafficking for debts incurred as part of their exploitation (54). Supporting this, Figures 4.6 and 4.7 map complaints geographically, showing that high-incidence areas correspond with known trafficking hot-spots (55).

These findings are consistent with academic insights into financial exploitation as a human trafficking tactic. Okoyeh and Kpalap (2023) describe modern debt bondage as a mechanism where victims are coerced into financial liabilities that are then weaponized against them (56). The complaints we analyzed align with this phenomenon-victims disputing debts they did not voluntarily incur, often involving repeated mentions of fraud, identity manipulation, or economic coercion.

This case study illustrates how unsupervised LLM-based tagging can identify emerging

regulatory issues without explicit labels, reflecting the broader phenomenon of emergent abilities observed in large language models (57). By surfacing themes like human trafficking, the model not only expands interpretability but also offers regulators and consumer protection agencies a new lens to detect financial harm. These insights could be used to support initiatives like the CFPB's 2024 enforcement actions against credit reporting violations involving vulnerable populations (58) or ongoing efforts to ensure fair treatment of human trafficking survivors (59; 60).

4.5 Temporal Trends and Potential

Although not part of the primary pipeline, temporal trends were detected. For example, time-series spikes in human trafficking-related tags may align with sociopolitical events such as the rollout of the Debt Bondage Repair Act. External datasets like the Global Slavery Index (61) or Polaris Project's human trafficking heatmaps (55) could further enhance such correlations in future work.

4.6 CHALLENGES AND LIMITATIONS

While the proposed system shows promise, challenges remain:

- Real-time processing at full dataset scale requires further infrastructure optimization.
- Clusters with high internal lexical variance (e.g., mixed categories)
- Need for better anomaly handling for vague or incoherent narratives
- LLM-generated labels sometimes generalized nuanced complaint differences.

Still, this pipeline is promising for deployment in consumer protection units or compliance departments, offering scalable categorization and early warning insights.

The findings demonstrate the inadequacy of manual hierarchical classification, especially at scale. By revealing hidden overlaps and ambiguous complaint categorizations, the semantic clustering + LLM labeling approach offers a dynamic, scalable alternative for modern

complaint management. The approach is not only adaptable to new complaint types but also enhances explainability (7; 5) through natural language theme generation.

CHAPTER 5: CONCLUSION AND FUTURE WORK

5.1 Summary of Findings

This thesis confirmed that consumer complaint narratives frequently transcend rigid hierarchical labels. Through semantic similarity analysis, clustering, and LLM-driven themes and tags, we demonstrate that a meaning-based complaint grouping approach is both feasible and superior in interpretability compared to traditional methods.

5.2 Conclusion

The manual categorization of complaints by static Product, Sub-product, Issue, and Sub-issue labels is increasingly inadequate in today's high-volume environment. This research shows that semantic embeddings, unsupervised clustering, and generative AI models can offer a scalable, narrative-aligned alternative. The proposed methods can significantly improve the efficiency of complaint resolution, regulatory precision, and consumer satisfaction.

5.3 Future Work

Future directions to extend this research include:

- Fine-tuning domain-specific LLMs (12) to enhance thematic summarization.
- Building real-time semantic clustering systems for the intake and triage of regulatory complaints in real time.
- Use tags to generate trends that can be used for other purposes
- Exploring Retrieval-Augmented Generation (RAG) pipelines using FAISS and GPT-4 to ground complaint explanations in regulatory text such as the Trafficking Victims

Protection Act (TVPA), enabling more accurate, explainable responses in legal and compliance contexts.

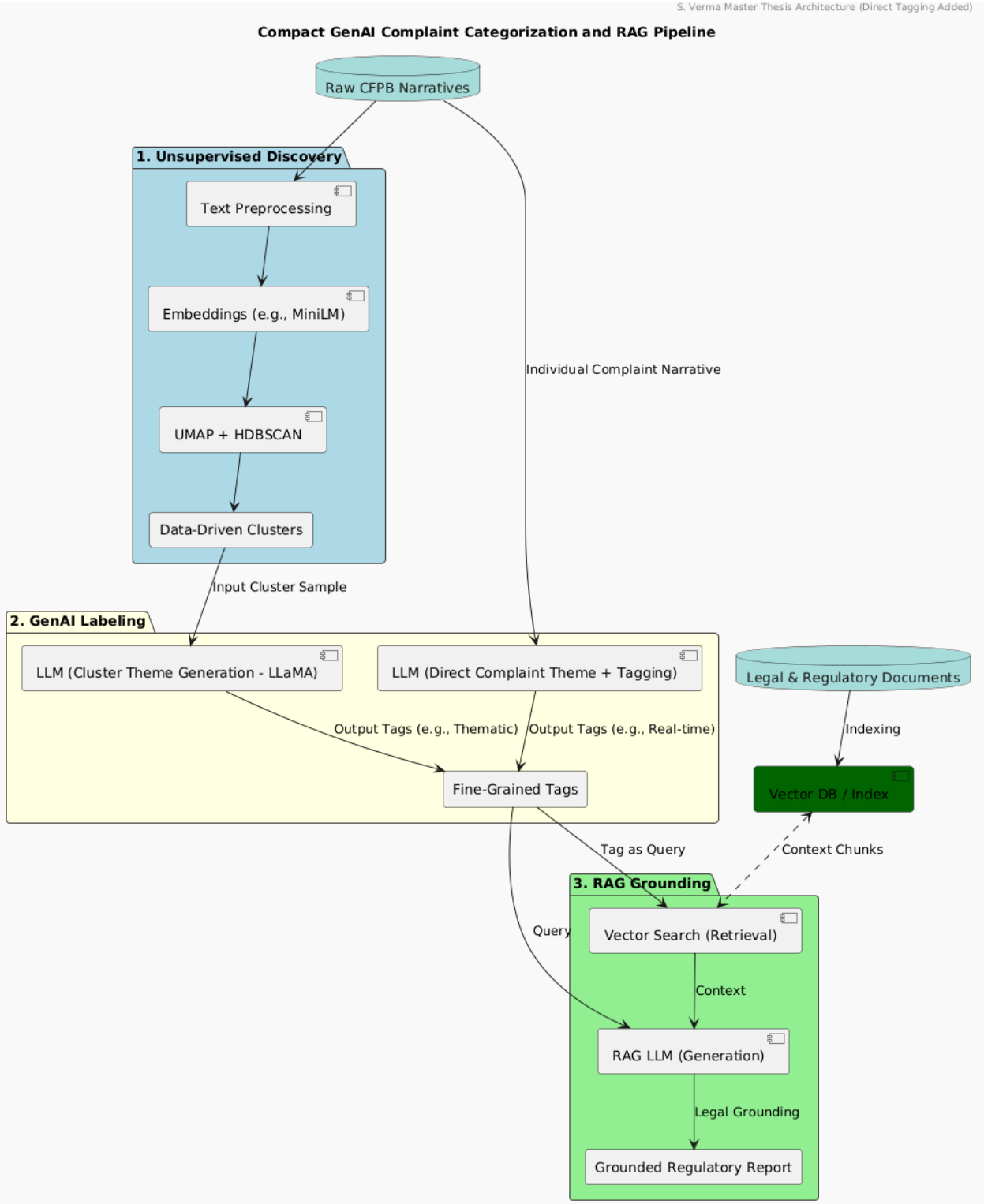


Figure 5.1: Proposed Pipeline that can be used for future work.

Future work may also explore methods for measuring computational surprise in complaint

narratives, building on recent approaches in emergent semantics (62).

5.4 Closing Remarks

Consumer complaints are dynamic, evolving narratives. A system built on semantic understanding rather than static categories has the potential to reshape regulatory reporting, improve corporate accountability, and better serve consumers in an increasingly complex financial landscape.

As newer LLMs like GPT-5 and Gemini 2.5 Flash emerge, future work could utilize them for theme coherence and hallucination resistance.

GitHub repository - containing the datasets and Jupyter notebooks is available here : **https://github.com/verma-sonal/ConsumerComplaints**

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APPENDIX A: Appendix

A.1 Selected Code Snippets

A.1.1 UMAP + HDBSCAN Clustering

```
umap_model = umap.UMAP(n_neighbors=15, min_dist=0.0, n_components=5)
umap_embeddings = umap_model.fit_transform(embeddings)
hdbscan_model = hdbscan.HDBSCAN(min_cluster_size=30)
cluster_labels = hdbscan_model.fit_predict(umap_embeddings)
```

A.1.2 LLaMA LLM Prompt Example

```
Prompt: "Given the following consumer complaints, summarize the
common theme across these complaints in 5-7 words."
Input: (10 complaint narratives)
Output: "Unauthorized Credit Card Charges"
```

A.2 Evaluation Metric Definitions

A.2.1 Silhouette Score

Silhouette Score $s(i)$ for a sample i is defined as:

$$s(i) = \frac{b(i) - a(i)}{\max\{a(i), b(i)\}}$$

where:

- $a(i)$ = average intra-cluster distance for i
- $b(i)$ = average nearest-cluster distance for i

A.2.2 Davies-Bouldin Index

The Davies-Bouldin Index (DBI) is calculated as:

$$DBI = \frac{1}{N} \sum_{i=1}^{N} \max_{j \neq i} \left(\frac{\sigma_i + \sigma_j}{d(c_i, c_j)} \right)$$

where:

- σ = average distance between each point in cluster and the cluster centroid.
- $d(c_i, c_j)$ = distance between cluster centroids.

A.3 Additional Figures

A.3.1 UMAP Cluster Scatter Plot

A.4 Sample Prompts and LLM Outputs

Example:

- **Cluster 101**
  - **Prompt Input:** 10 sampled complaints about credit card late fees.
  - **LLM Output:** "Disputes on Late Payment Charges"
- **Cluster 205**
  - **Prompt Input:** 10 sampled complaints about mortgage servicing.
  - **LLM Output:** "Mortgage Account Errors and Payment Issues"

A.5 Environment and Libraries

- Python 3.10
- SentenceTransformers (HuggingFace)
- UMAP-learn
- HDBSCAN
- LLaMA-3.2-3B-Instruct
- Hardware: 64GB RAM, NVIDIA T4 GPU (Colab Pro+)